# GROND Workflow Figures

Global Regularization Of Non-kinematical Diffraction

## Figure Set

These figures describe how GROND takes a reintegrated XDS directory plus a solved structure and turns row-level observations into Bloch/Born predictions, corrected reflection files, endpoint statistics, forensic audits, and maps. The diagrams are written as source figures for reports, talks, and lab documentation. For a non-expert reader, a *row* means one measured spot observation from the integration table. A *merged reflection* means several rows have already been combined. GROND keeps rows alive for as long as possible because row-level geometry, thickness, hand, and weak-reflection behavior are where many useful diagnostics live.

## Overall GROND Operating Map

The overview is intentionally compact: solve once, reintegrate for row-level observations, then let GROND make staged decisions before final outputs.

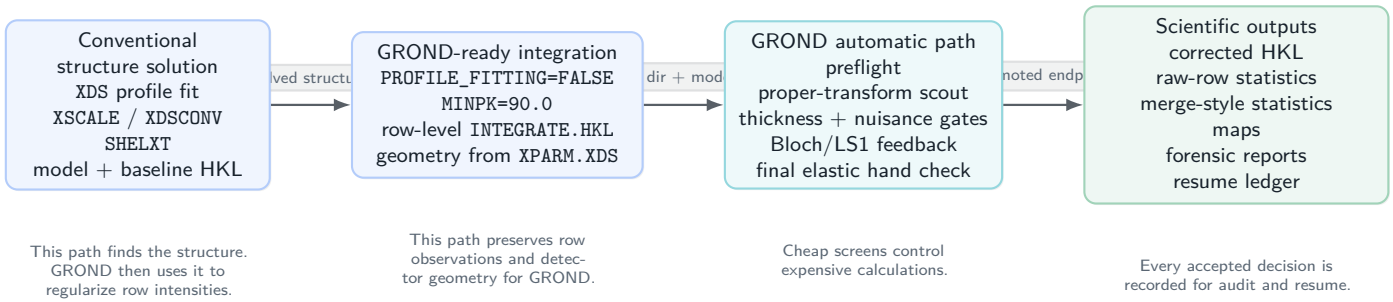



**Figure 1:** High-level GROND flow. The conventional path supplies a chemically sensible starting model; the GROND path starts from row-level integration data and regularizes observed intensities using non-kinematical Bloch/Born predictions.

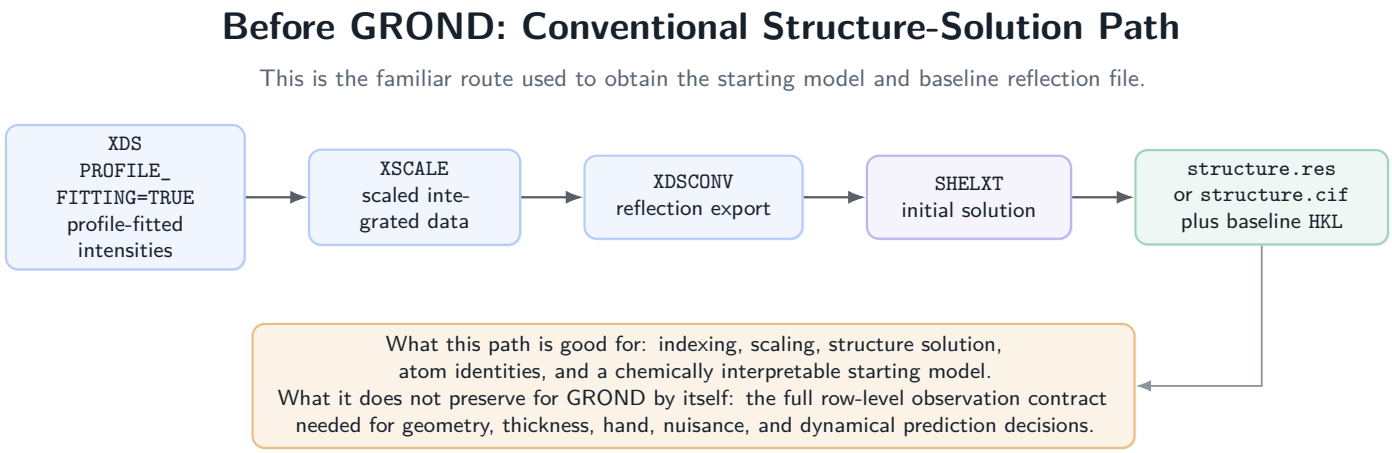



**Figure 2:** The pre-GROND path produces the starting model and baseline HKL. GROND keeps this path as provenance but does not treat it as the final physical model.

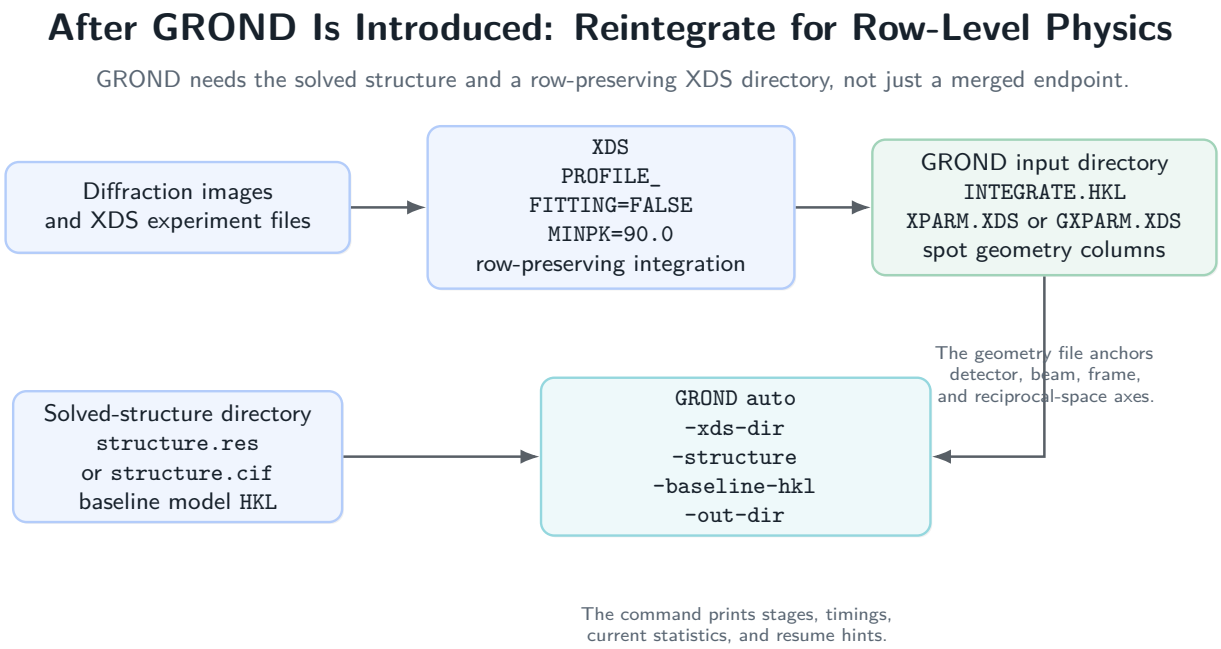



**Figure 3:** Recommended GROND intake. The row-preserving XDS run provides the observation contract; the solved structure provides the atomic model and baseline reflection alignment.

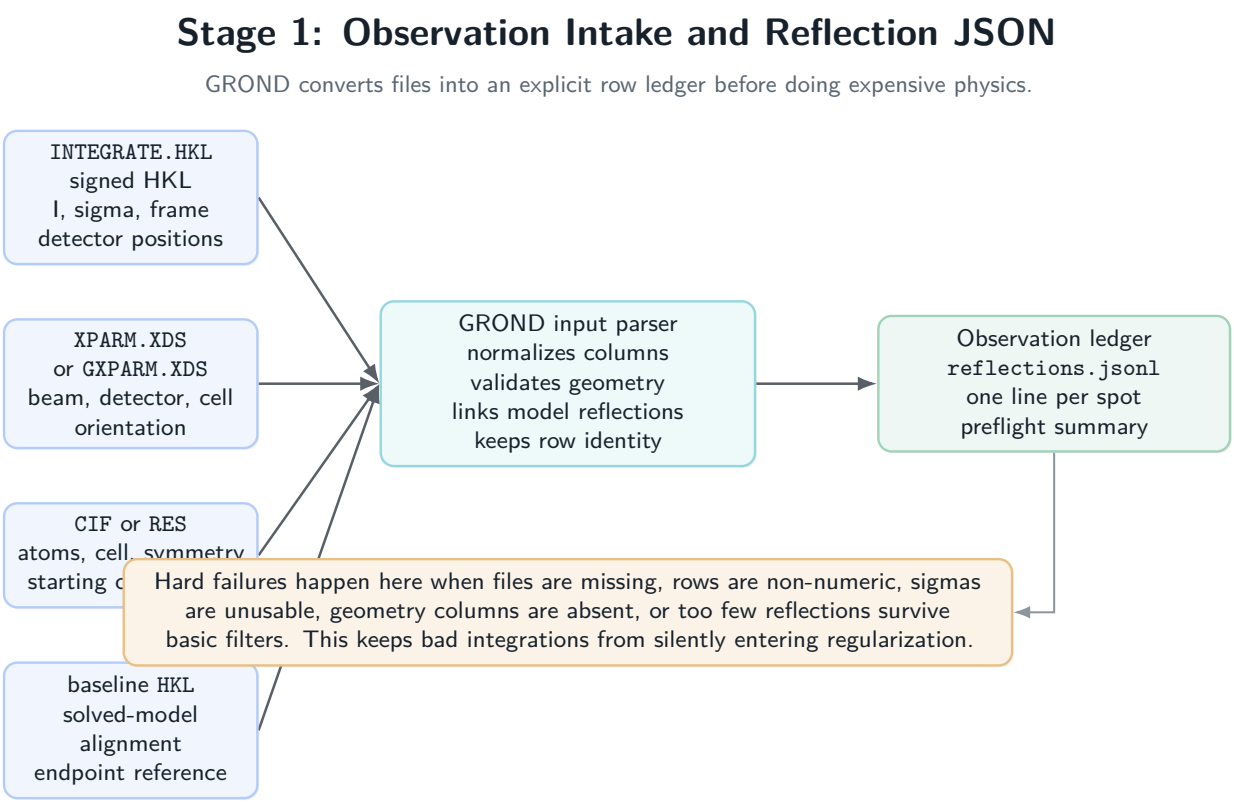



**Figure 4:** Reflection JSON creation. GROND first builds a machine-readable observation ledger so later stages can be audited, resumed, and compared row by row.

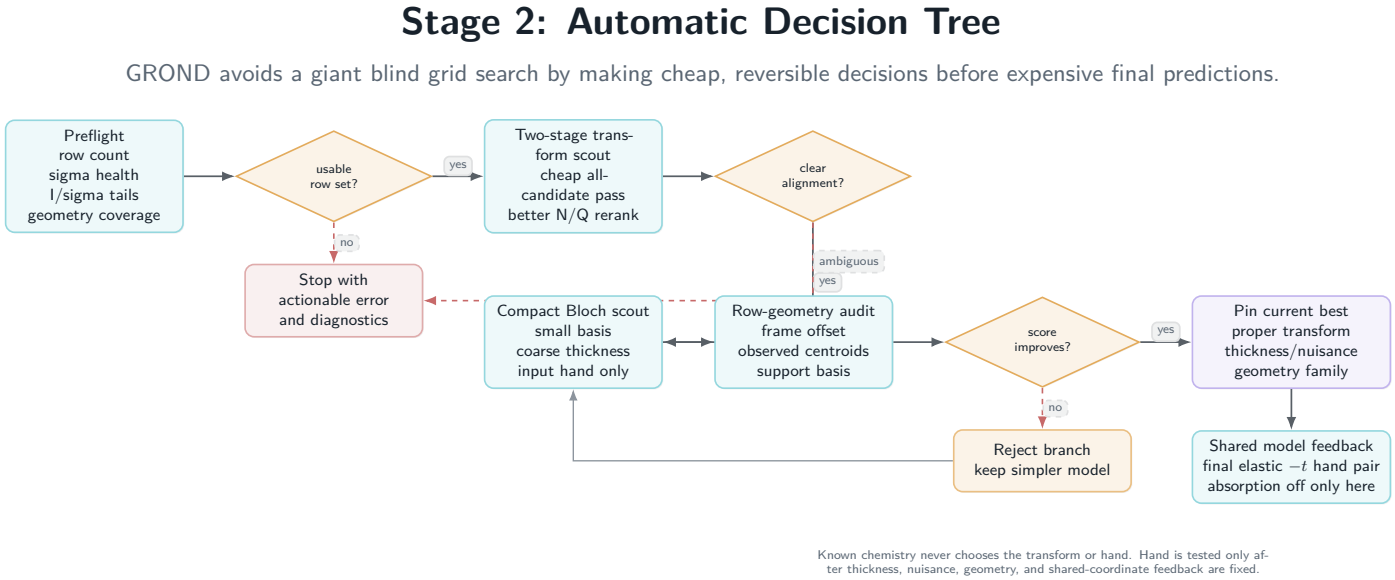



**Figure 5:** Automatic decision tree. Chemistry-blind proper-transform selection is separated from hand. The conditional hand comparison occurs at the end using the exact elastic negative-thickness identity with final rows and nuisance terms fixed.

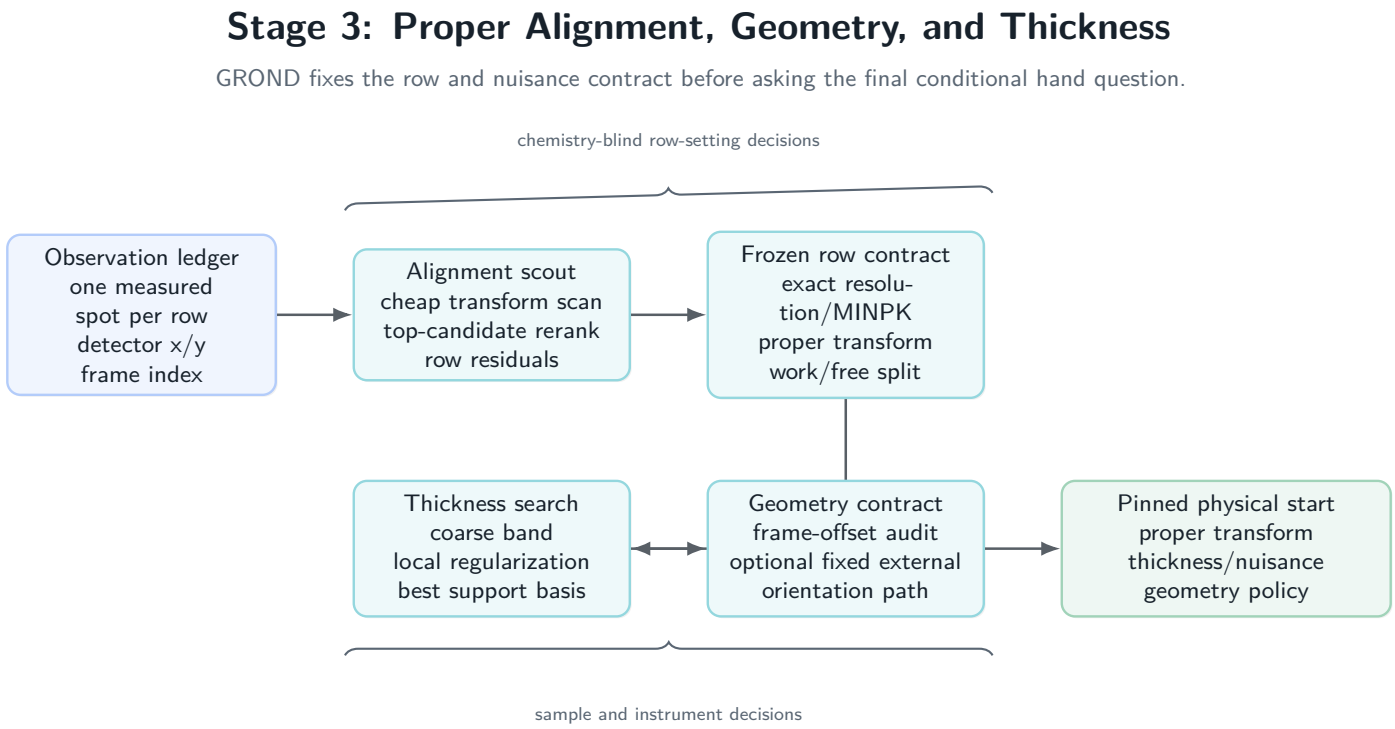



**Figure 6:** Early physical decisions. The goal is to fix a stable determinant-positive transform, row support, thickness band, and geometry/nuisance family before coordinate feedback. Hand is deliberately absent from this stage.

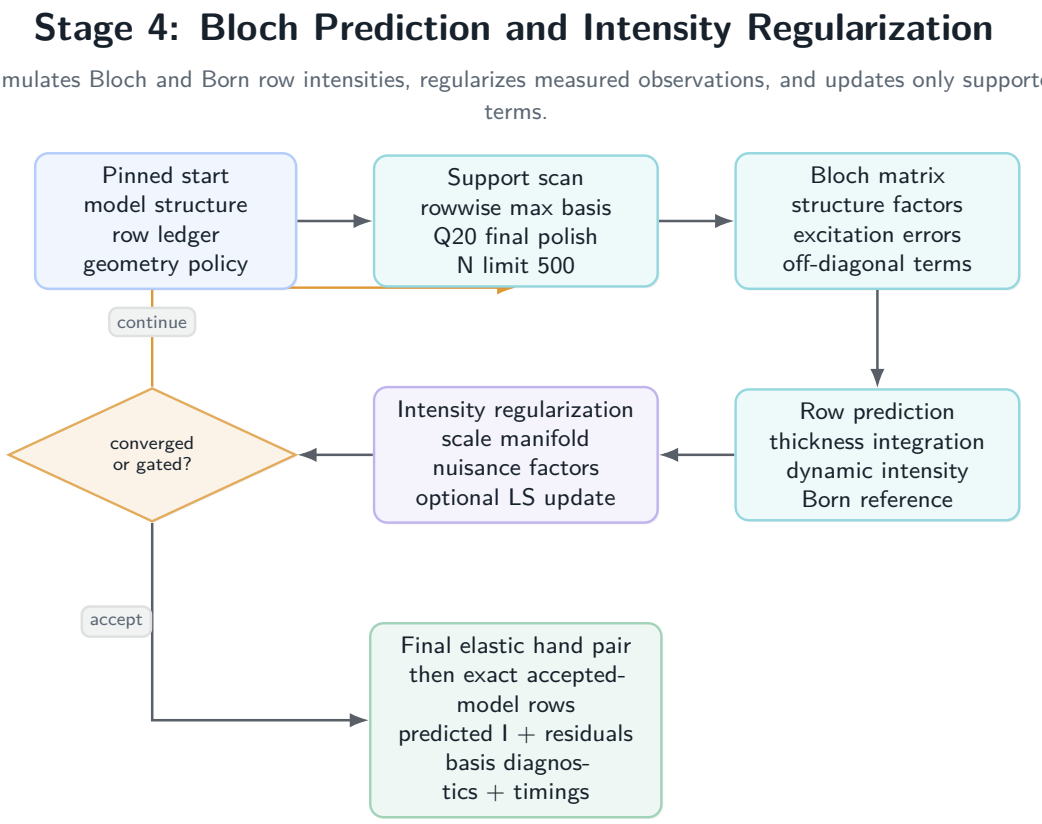



**Figure 7:** Bloch/regularization core. After coordinate feedback converges or stops, GROND performs one matched elastic negative-thickness hand comparison. If the hand changes, the accepted model is materialized and its rows are recomputed without another coordinate step.

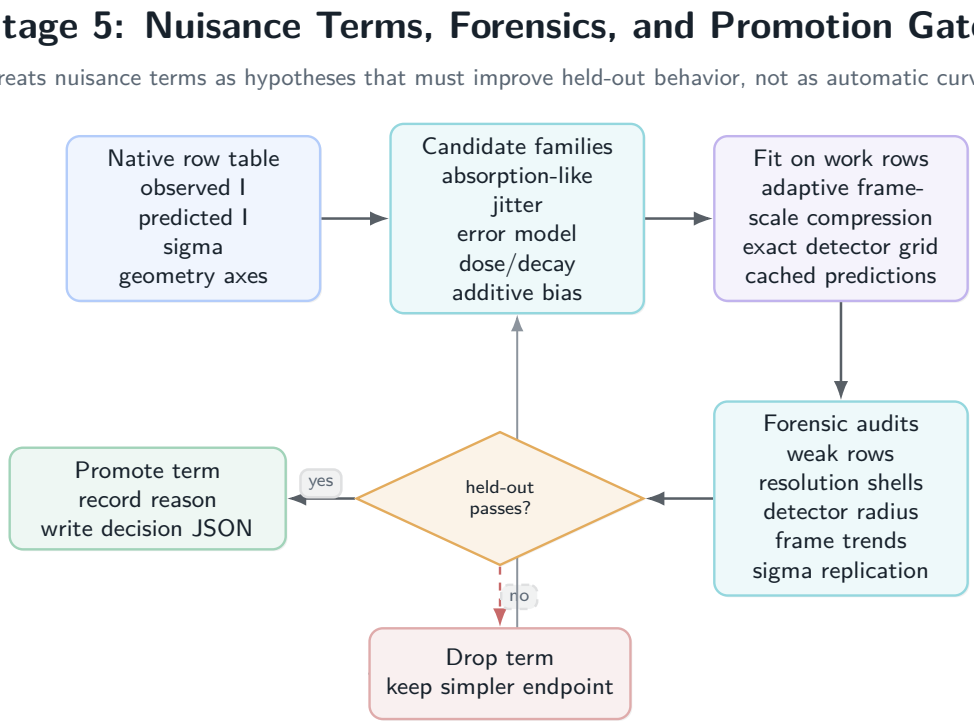



**Figure 8:** Nuisance and forensic gatekeeping. The package promotes only terms that improve the physical model without hiding failures in scale factors or weak reflections.

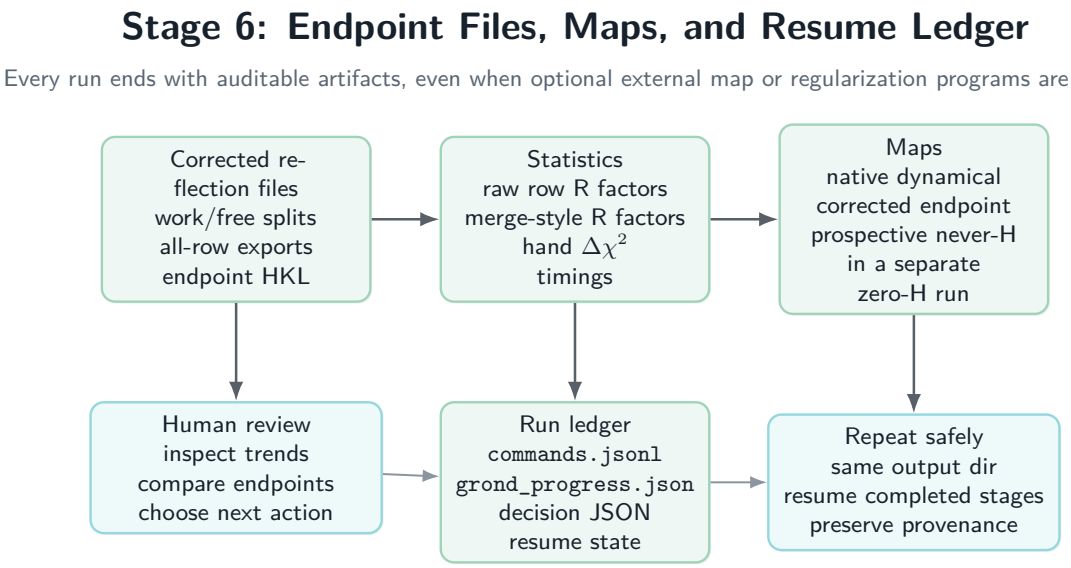



**Figure 9:** Endpoint and artifact production. GROND produces corrected reflection files, statistics, maps when possible, and a resume ledger so interrupted runs can continue without losing provenance.

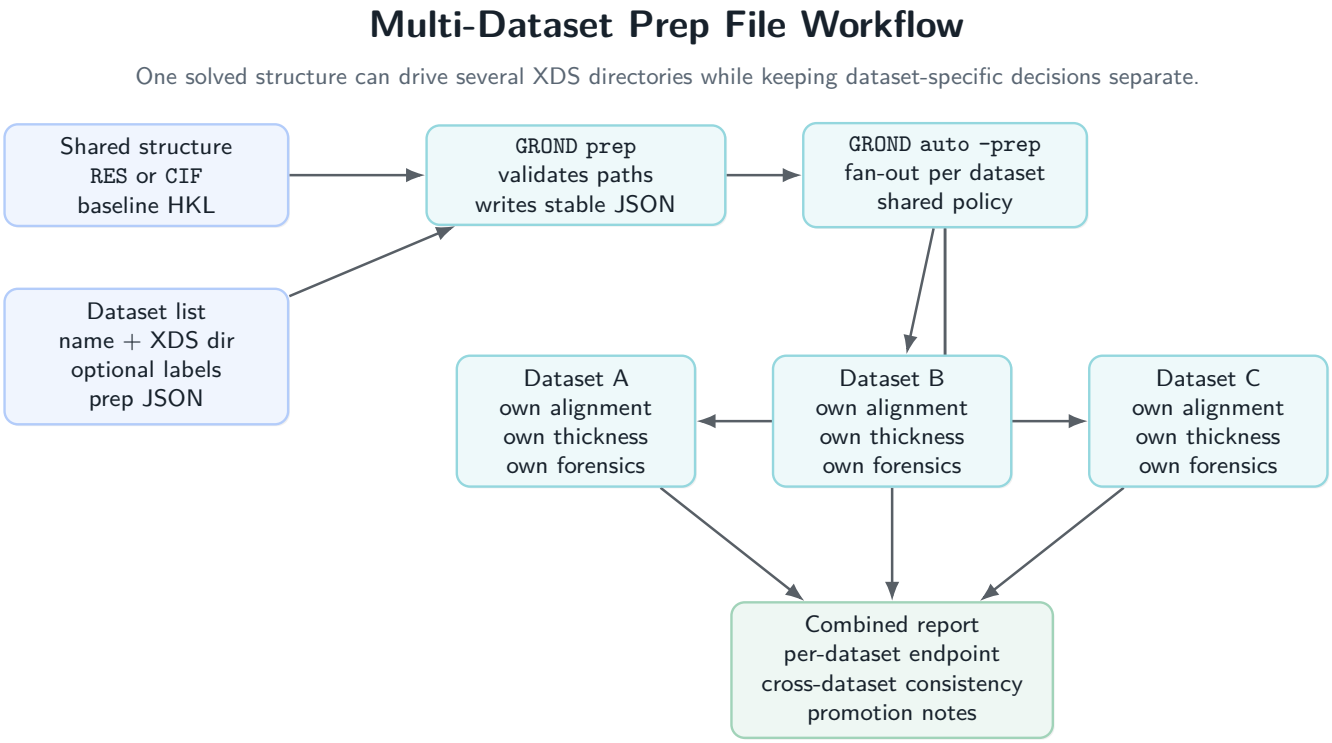



**Figure 10:** Multi-dataset execution. The prep file is a compact manifest: shared structure and baseline model information plus multiple row-preserving XDS directories.

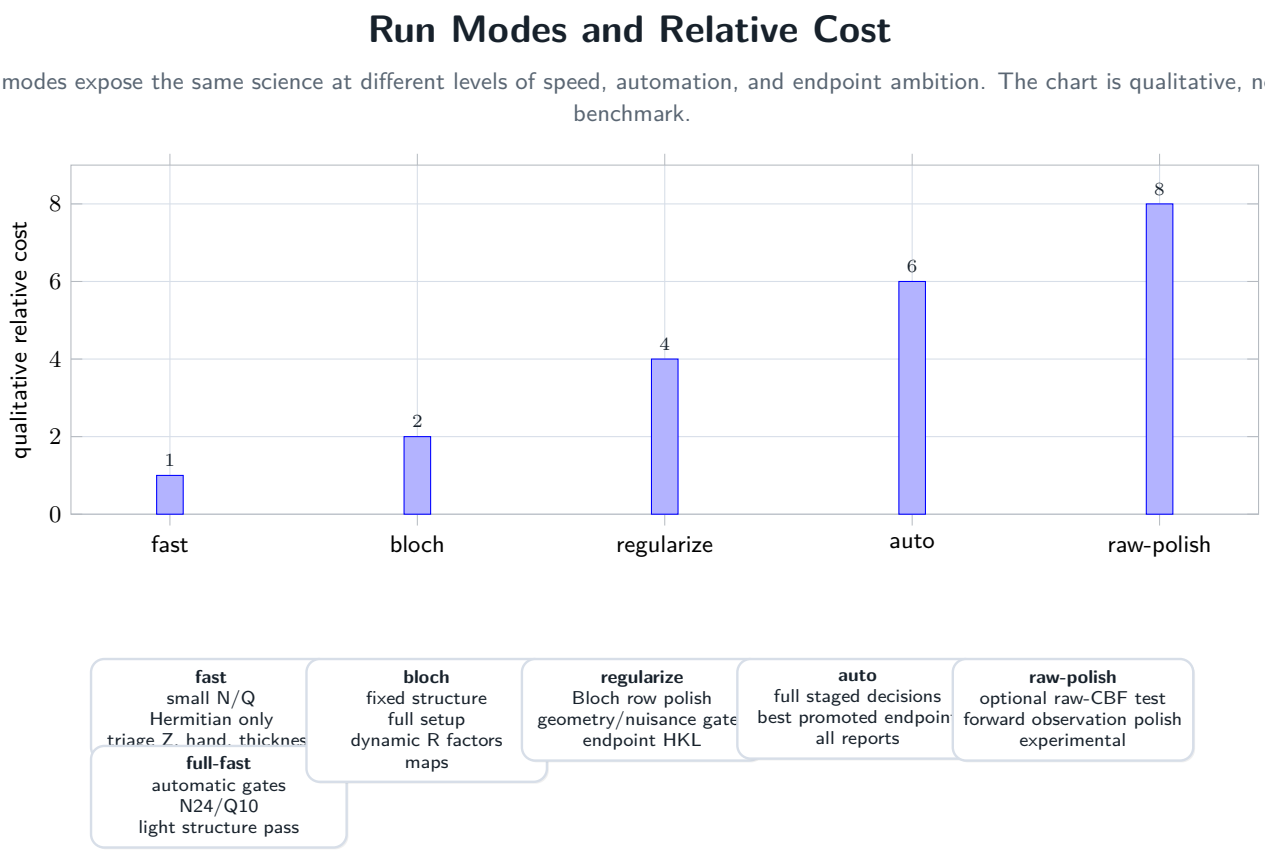



**Figure 11:** Mode chart. Fast mode is for immediate sanity checks; auto mode is the default walk-away path; raw-polish remains an optional final test layer when raw images and calibration are available.